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COS 442

HW5

4/13/2025

Overview

This file contains Python source code, docker files, a docker-compose file, and an Nginx config file to build and run four small micros servers and a reverse proxy on Nginx, as well as screenshots of the results.

What you need

Python 3

Docker

Nginx

Files Included

Wind Folder: This folder contains the wind.py file, the Python source code, a Docker file, and a requirements.txt file to build and run the wind container.

Temp Folder: This folder contains the temp.py file, the Python source code, a Docker file, and a requirements.txt file to build and run the temp container.

Humidity Folder: This folder contains the humidity.py file, the Python source code, a Docker file, and a requirements.txt file to build and run the humidity container.

Weatherman Folder: This folder contains the weatherman .py file, the Python source code, a Docker file, and a requirements.txt file to build and run the weatherman container.

nginx.conf: This file is the config file to run the nginx reverse proxy

Docker-compose.yaml: docker compose file that runs and sets up all four containers.

my_conf: This is a text file that is a copy of the nginx.conf file.

4 screenshots showing the results

Resources

I used some of the code from the IC4 and HW5 slides to help as my boilerplate for my micro services and my nginx.conf file

For help with Flask, I used this article (https://flask.palletsprojects.com/en/stable/quickstart/)

For help with Python Request, I used this article (https://www.w3schools.com/python/module_requests.asp)

For help with Python Random, I used this article (https://www.w3schools.com/python/module_random.asp)